

Sushant Aryal

📍 Louvain-la-Neuve, Belgium 🌐 www.aryalsushant.com.np ☎ +32 496 02 65 57 ✉ aryalsushant564@gmail.com

• ABOUT ME

MSc graduate in Geo-Information Science and Earth Observation, specialized in fusing multimodal, multitemporal, and multiresolution satellite imagery with machine learning to model complex environmental and spatiotemporal dynamics. Experienced in handling large-scale remote sensing datasets and building reproducible geospatial workflows. Motivated to advance physics-informed machine learning and geospatial foundation models for scalable, Earth system monitoring.

• RESEARCH INTEREST

Remote sensing and spatial data science for environmental monitoring; multi-sensor, multitemporal satellite data analysis; physics-informed machine learning and geospatial foundation models; geospatial analysis and programming for sustainable environmental decision-making

• EDUCATION

Sep 2024 – Jun 2026

Geo-Information Science and Earth Observation for Environmental Modelling and Management (MSc)

University of Tartu, Estonia · Université catholique de Louvain, Belgium · GPA: 4.38/5.00 (Year 1), 16.19/20 (Year 2)

Coursework: Data Science in Remote Sensing, Land Monitoring by Advanced Satellite Remote Sensing, Spatial Modelling of Land Dynamics, Advanced Geo-processing, Geospatial Analysis with Python and R, Applied Geomatics, spatial data studio

Thesis: Comparative Evaluation of Sentinel-1, Sentinel-2, and Fused Time Series for Agricultural Field Boundary Delineation Using Deep Learning

Supervisors: Prof. Pierre Defourny, Quentin Deffense

- Adapted PTAViT3D, a 3D Vision Transformer leveraging joint spatio-temporal attention, for multi-task agricultural field boundary delineation
- Extended the framework with Sentinel-1 interferometric coherence alongside backscatter, exploiting phenology-sensitive SAR features for crop boundary discrimination
- Compared single-sensor (S2 cloud-free, S2 partially cloudy, S1 backscatter + coherence) and fused S1/S2 cross-attention configurations against reference datasets

Aug 2018 – May 2023

Geomatics Engineering (BSc)

Kathmandu University, Dhulikhel, Nepal · GPA: 3.71/4.00

Coursework: Remote Sensing, Geographic Information System, Python Programming, Spatial Database Management, Photogrammetry, Numerical Methods

Thesis: A WebGIS-Based Spatial Decision Support System for Disaster Management: A Case Study of Lalitpur Metropolitan City

• SKILLS

Geospatial Data Science & GIS: GeoPandas, Rasterio, GDAL, PostgreSQL/PostGIS, spatial SQL, QGIS, ArcGIS Pro

Remote Sensing & Earth Observation: Sentinel-1 SAR (coherence, backscatter), Sentinel-2 optical analysis, multitemporal image analysis, orthophoto, UAV

Machine Learning & Deep Learning: Neural networks, Computer Vision, pattern recognition, CNNs, U-Net and segmentation models, Vision Transformers, multi-task learning, model training & evaluation, PyTorch, TensorFlow, scikit-learn

Programming & Reproducible Workflows: Python, JavaScript, MATLAB, R, Git, large scale data handling (HDF5, Zarr), reproducible pipeline design

High-Performance Computing: Linux / SLURM environments, GPU-based model training

Collaboration & Communication: teamwork in multicultural and interdisciplinary settings, adaptability, problem-solving, attention to detail, conference/seminar presentation, technical report and documentation writing

● EXPERIENCE

Student Assistant

June 2026 – July 2026

Earth and Life Institute, Université catholique de Louvain – Louvain-la-Neuve, Belgium

- Applied field boundary delineation model developed in my thesis to generate 2026 agricultural field boundary maps for the Wallonia region
- Processed and fused Sentinel-2 time series data for the Walloon agricultural landscape to support regional-scale delineation

Geospatial AI Research Internship

May 2025 – Aug 2025

Estonian Land and Spatial Development Board – Tartu, Estonia

- Built a U-Net segmentation pipeline with a ResNet backbone for solar panel detection across aerial imagery, including spatial accuracy assessment of model predictions against ground reference data.
- Researched building height estimation from oblique imagery, using heatmap-based window detection as a structural cue for floor-count inference.
- Designed reproducible Python geospatial pipelines (PyTorch, GDAL, Rasterio, NumPy) for large-scale raster processing across national aerial imagery datasets.

GIS Developer

May 2024 – Aug 2024

PlantSat – Kathmandu, Nepal

- Built server-side geospatial pipelines for a satellite-based agricultural monitoring platform, integrating satellite data, vegetation indices, and field-level analytics.
- Managed and processed vector and raster geospatial data using PostGIS, GDAL, and GeoPandas, including database design and spatial indexing.
- Developed GIS data services and APIs with geometry validation, spatial filtering, and export functionality for large-scale geospatial datasets.

● PROJECTS

Spatio-Temporal Wildfire Prediction in California using Multi-Source Earth Observation Data

Course Project, Data Science in Bioscience Engineering, UCLouvain, 2025

GitHub: github.com/SushantAryal3/Deep-learning-Wild-Fire-Prediction-

- Built spatio-temporal deep learning pipeline for next-day wildfire prediction across California, integrating static (terrain, infrastructure), dynamic (10-day meteorological time series), and land-cover variables
- Adapted the open-source LOAN (Location-Aware Adaptive Normalization) architecture for the California dataset, with 3D convolutions for meteorological sequences and static feature modulation

● SCHOLARSHIPS/AWARDS

Erasmus Mundus Joint Master Degree Scholarship (2024–2026)

Competitive EU scholarship for the GEM MSc programme in Geo-Information Science and Earth Observation, Track 2: Land Use Systems — full tuition, stipend, and travel allowance, total value €48,000

Nepal Government Scholarship (2018–2023)

Merit-based 35% tuition scholarship for undergraduate studies, Kathmandu University — approx. USD 2,270

● ONLINE PROFILES

LinkedIn: [linkedin.com/in/sushant-aryal-56360b1a4](https://www.linkedin.com/in/sushant-aryal-56360b1a4)

GitHub: github.com/SushantAryal3

ORCID: orcid.org/0009-0002-6461-1858

● LANGUAGES

English (Proficient C1), French (A2), Dutch (A2), Nepali (fluent)